

Plan Switch Correction Analysis

Summary of the results from Task 03:

Query	Switch	Classification	Search Direction
Q1	S2	DELAYED	Search BEFORE 1992-01-27
Q1	S3	EARLY	Search AFTER 1992-03-30
Q2	S1	EARLY	Search AFTER 1992-01-02
Q2	S2	EARLY	Search AFTER 1992-05-14
Q3	S1	EARLY	Search AFTER 1992-01-02
Q3	S2	DELAYED	Search BEFORE 1992-01-03
Q3	S3	EARLY	Search AFTER 1992-01-07
Q3	S4	EARLY	Search AFTER 1992-01-19
Q3	S6	EARLY	Search AFTER 1992-03-09
Q3	S7	DELAYED	Search BEFORE 1992-03-13

CORRECT switches (skip): Q1 S1, Q2 S3, Q3 S5, Q3 S8

1. Query 1 Analysis:

Switch S2 (DELAYED):

- **Optimizer's Switch:** 1992-01-27 → 1992-01-28
- **Optimal Switch:** 1992-01-09 → 1992-01-10
- **Execution Times at Optimal Switch:**

$RT(P_i, q_i') = 24.43 \text{ ms}$

$RT(P_j, q_j') = 25.30 \text{ ms}$

$RT(P_i, q_j') = 40.10 \text{ ms (forced)}$

$RT(P_j, q_i') = 27.86 \text{ ms (forced)}$

- **Explanation:**

The optimizer delayed this switch by more than two weeks. Testing reveals that the optimal crossover point occurred between Jan 9 and Jan 10 where Pi was legitimately faster on Jan 9 (24.43 ms < 27.86 ms) and Pj became faster on Jan 10 (25.30 ms < 40.10 ms).

Switch S3 (EARLY):

- **Optimizer's Switch:** 1992-03-29 → 1992-03-30
- **Optimal Switch:** 1998-02-21 → 1998-02-22 (Approximate / Boundary Case)
- **Execution Times at Optimal Switch:**

$RT(P_i, q_i') = 21051.96 \text{ ms}$

$RT(P_j, q_j') = 20018.65 \text{ ms}$

$RT(P_i, q_j') = 20056.97 \text{ ms (forced)}$

$RT(P_j, q_i') = 19926.20 \text{ ms (forced)}$

- **Explanation:**

The optimizer switched to Pj in March 1992, which was drastically early. Pi remained the dominant plan for nearly six years.

Binary search isolated the crossover to late February 1998. The slight validation failure at q_i' (where Pj was marginally faster by ~1 second) indicates that performance was highly volatile around this specific date due to data distribution.

However, the macro-level conclusion remains clear: the optimizer abandoned Pi years before it should have.

2. Query 2 Analysis:

Switch S1 (EARLY):

- **Optimizer's Switch:** 1992-01-01 → 1992-01-02
- **Optimal Switch:** NOT FOUND (No crossover in data range)
- **Explanation:**

Performance testing across the entire date range (up to 1998-08-02) shows that Pi strictly dominates Pj.

The execution time of Pi remains consistently lower than Pj throughout the dataset.

Therefore, the optimizer made a completely flawed decision on day 2 and should never have abandoned Pi.

Switch S2 (EARLY):

- **Optimizer's Switch:** 1992-05-13 → 1992-05-14
- **Optimal Switch:** 1997-02-01 → 1997-02-02
- **Execution Times at Optimal Switch**

$RT(P_i, q_i') = 21267.14$ ms

$RT(P_j, q_j') = 19558.80$ ms

$RT(P_i, q_j') = 24367.67$ ms (forced)

$RT(P_j, q_i') = 26040.85$ ms (forced)

- **Explanation**

The optimizer switched to Pj in May 1992, but the true optimal crossover point occurred almost five years later in February 1997. The validation cleanly passes, confirming Pi was better on Feb 1 and Pj became better on Feb 2.

3. Query 3 Analysis:

Switch S1 (EARLY):

- **Optimizer's Switch:** 1992-01-01 → 1992-01-02
- **Optimal Switch:** 1996-12-04 → 1996-12-05
- **Execution Times at Optimal Switch**

$RT(P_i, q_i') = 16007.11$ ms

$RT(P_j, q_j') = 15926.46$ ms

$RT(P_i, q_j') = 16470.02$ ms (forced)

$RT(P_j, q_i') = 19536.78$ ms (forced)

- **Explanation**

The optimizer prematurely dropped Pi on the second day of the dataset. Testing confirms that Pi remained optimal until December 1996, when Pj finally overtook it.

Switch S2 (DELAYED):

- **Optimizer's Switch:** 1992-01-03 → 1992-01-04
- **Optimal Switch:** 1992-01-01 → 1992-01-02 (Boundary Case)
- **Execution Times at Optimal Switch**

$RT(P_i, q_i') = 0.02 \text{ ms}$
 $RT(P_j, q_j') = 102.43 \text{ ms}$
 $RT(P_i, q_j') = 96.76 \text{ ms (forced)}$
 $RT(P_j, q_i') = 0.02 \text{ ms (forced)}$

- **Special Scenario Explanation**

This is a boundary case at the very start of the dataset. Because the date filter is extremely restrictive on 1992-01-01, both plans effectively return empty sets instantly (~0.02 ms). However, by 1992-01-02, P_j was already intended to be the superior plan. The optimizer delayed switching until Jan 4.

Switch S3 (EARLY):

- **Optimizer's Switch:** 1992-01-06 → 1992-01-07
- **Optimal Switch:** 1995-06-13 → 1995-06-14
- **Execution Times at Optimal Switch:**

$RT(P_i, q_i') = 17492.56 \text{ ms}$
 $RT(P_j, q_j') = 16864.17 \text{ ms}$
 $RT(P_i, q_j') = 17275.39 \text{ ms (forced)}$
 $RT(P_j, q_i') = 22370.94 \text{ ms (forced)}$

- **Explanation:**

The optimizer switched on Jan 7, 1992, but the optimal crossover occurred over three years later in June 1995. The validation clearly confirms this crossing point.

Switch S4 (EARLY):

- **Optimizer's Switch:** 1992-01-18 → 1992-01-19
- **Optimal Switch:** NOT FOUND (No crossover in data range)
- **Explanation:**

Similar to Query 2 (S_1), the optimizer switched prematurely. Testing across the full dataset (up to 1998-08-02) shows that the original plan P_i remained consistently better. Therefore, the optimizer should never have switched plans.

Switch S6 (EARLY):

- **Optimizer's Switch:** 1992-03-08 → 1992-03-09
- **Optimal Switch:** NOT FOUND (No crossover in data range)
- **Explanation:**

Again, Pi remained the dominant plan throughout the entire dataset. The optimizer's decision to switch to Pj on 1992-03-09 was incorrect, and a crossover never occurred.

Switch S₇ (DELAYED)

- **Optimizer's Switch:** 1992-03-13 → 1992-03-14
- **Optimal Switch:** 1992-01-01 → 1992-01-02 (Boundary Case)
- **Execution Times at Optimal Switch:**

RT(Pi, qi') = 0.01 ms

RT(Pj, qj') = 152.59 ms

RT(Pi, qj') = 100.60 ms (forced)

RT(Pj, qi') = 0.02 ms (forced)

- **Explanation:**

The optimizer held onto Pi until March 1992, but performance testing shows Pj was already better from the beginning of the dataset. On 1992-01-02, the forced execution of Pi: 100.60 ms was already significantly slower than Pj: 152.59 ms. Therefore, the optimizer should have switched immediately.

Terminal Output on running [task.py](#) from 03:

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Query 1 — Switch S2 (DELAYED)
Optimizer's switch: 1992-01-27 → 1992-01-28
=====
Searching for optimal switch...
Boundary [1992-01-02]: RT(Pi)=0.9ms RT(Pj)=1.1ms
[1992-01-02 ... 1992-01-14 ... 1992-01-27] RT(Pi)=387.5 RT(Pj)=68.3 → Pj better, search
LEFT
[1992-01-02 ... 1992-01-08 ... 1992-01-14] RT(Pi)=56.4 RT(Pj)=57.0 → Pi better, search
RIGHT
[1992-01-08 ... 1992-01-11 ... 1992-01-14] RT(Pi)=91.7 RT(Pj)=51.7 → Pj better, search
LEFT
[1992-01-08 ... 1992-01-09 ... 1992-01-11] RT(Pi)=22.8 RT(Pj)=24.5 → Pi better, search
RIGHT
[1992-01-09 ... 1992-01-10 ... 1992-01-11] RT(Pi)=39.2 RT(Pj)=29.2 → Pj better, search
LEFT

Optimal switch found: 1992-01-09 → 1992-01-10
Validating...
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Measuring RT(Pi, d*i=1992-01-09)... 24.43 ms
Measuring RT(Pj, d*j=1992-01-10)... 25.30 ms
Measuring RT(Pi, d*j=1992-01-10) [forced]... 40.10 ms
Measuring RT(Pj, d*i=1992-01-09) [forced]... 27.86 ms

Pi better before: True (24.4 vs 27.9)
Pj better after: True (25.3 vs 40.1)
Valid optimal switch: YES

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Query 1 — Switch S3 (EARLY)
Optimizer's switch: 1992-03-29 → 1992-03-30
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Searching for optimal switch...
Boundary [1998-12-01]: RT(Pi)=34857.0ms RT(Pj)=27764.1ms
[1992-03-30 ... 1995-07-31 ... 1998-12-01] RT(Pi)=12655.3 RT(Pj)=14999.5 → Pi better,
search RIGHT
[1995-07-31 ... 1997-03-31 ... 1998-12-01] RT(Pi)=16480.0 RT(Pj)=17215.8 → Pi better,
search RIGHT
[1997-03-31 ... 1998-01-30 ... 1998-12-01] RT(Pi)=19386.1 RT(Pj)=26635.1 → Pi better,
search RIGHT
[1998-01-30 ... 1998-07-01 ... 1998-12-01] RT(Pi)=80024.0 RT(Pj)=19952.0 → Pj better,
search LEFT
[1998-01-30 ... 1998-04-16 ... 1998-07-01] RT(Pi)=35883.8 RT(Pj)=23074.8 → Pj better,
search LEFT
[1998-01-30 ... 1998-03-09 ... 1998-04-16] RT(Pi)=36398.8 RT(Pj)=27604.7 → Pj better,
search LEFT
[1998-01-30 ... 1998-02-18 ... 1998-03-09] RT(Pi)=23686.4 RT(Pj)=27091.8 → Pi better,
search RIGHT
[1998-02-18 ... 1998-02-27 ... 1998-03-09] RT(Pi)=29535.2 RT(Pj)=29307.7 → Pj better,
search LEFT
[1998-02-18 ... 1998-02-22 ... 1998-02-27] RT(Pi)=24248.9 RT(Pj)=22793.3 → Pj better,
search LEFT
[1998-02-18 ... 1998-02-20 ... 1998-02-22] RT(Pi)=19796.5 RT(Pj)=20116.3 → Pi better,
search RIGHT
[1998-02-20 ... 1998-02-21 ... 1998-02-22] RT(Pi)=19792.2 RT(Pj)=19984.4 → Pi better,
search RIGHT

Optimal switch found: 1998-02-21 → 1998-02-22
Validating...
Measuring RT(Pi, d*i=1998-02-21)... 21051.96 ms
Measuring RT(Pj, d*j=1998-02-22)... 20018.65 ms
Measuring RT(Pi, d*j=1998-02-22) [forced]... 20056.97 ms
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Measuring RT(Pj, d*i=1998-02-21) [forced]... 19926.20 ms

Pi better before: False (21052.0 vs 19926.2)

Pj better after: True (20018.7 vs 20057.0)

Valid optimal switch: NO (boundary case)

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Query 2 — Switch S1 (EARLY)

Optimizer's switch: 1992-01-01 → 1992-01-02

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Searching for optimal switch...

Boundary [1998-08-02]: RT(Pi)=32529.6ms RT(Pj)=33432.9ms

Pi always better in range — no crossover found

No crossover found — Pi remains better throughout range

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Query 2 — Switch S2 (EARLY)

Optimizer's switch: 1992-05-13 → 1992-05-14

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Searching for optimal switch...

Boundary [1998-08-02]: RT(Pi)=33950.9ms RT(Pj)=32106.5ms

[1992-05-14 ... 1995-06-23 ... 1998-08-02] RT(Pi)=32780.6 RT(Pj)=35976.3 → Pi better,
search RIGHT

[1995-06-23 ... 1997-01-11 ... 1998-08-02] RT(Pi)=35285.9 RT(Pj)=36281.6 → Pi better,
search RIGHT

[1997-01-11 ... 1997-10-22 ... 1998-08-02] RT(Pi)=34992.7 RT(Pj)=34823.9 → Pj better,
search LEFT

[1997-01-11 ... 1997-06-02 ... 1997-10-22] RT(Pi)=34024.3 RT(Pj)=32990.8 → Pj better,
search LEFT

[1997-01-11 ... 1997-03-23 ... 1997-06-02] RT(Pi)=32969.6 RT(Pj)=32907.8 → Pj better,
search LEFT

[1997-01-11 ... 1997-02-15 ... 1997-03-23] RT(Pi)=34251.8 RT(Pj)=33304.8 → Pj better,
search LEFT

[1997-01-11 ... 1997-01-28 ... 1997-02-15] RT(Pi)=33624.2 RT(Pj)=36596.3 → Pi better,
search RIGHT

[1997-01-28 ... 1997-02-06 ... 1997-02-15] RT(Pi)=36699.3 RT(Pj)=22106.4 → Pj better,
search LEFT

[1997-01-28 ... 1997-02-01 ... 1997-02-06] RT(Pi)=20249.1 RT(Pj)=20699.0 → Pi better,
search RIGHT

[1997-02-01 ... 1997-02-03 ... 1997-02-06] RT(Pi)=19901.3 RT(Pj)=19698.3 → Pj better,
search LEFT

[1997-02-01 ... 1997-02-02 ... 1997-02-03] RT(Pi)=19507.2 RT(Pj)=19084.4 → Pj better,
search LEFT

Optimal switch found: 1997-02-01 → 1997-02-02

Validating...

Measuring RT(Pi, d*i=1997-02-01)... 21267.14 ms

Measuring RT(Pj, d*j=1997-02-02)... 19558.80 ms

Measuring RT(Pi, d*j=1997-02-02) [forced]... 24367.67 ms

Measuring RT(Pj, d*i=1997-02-01) [forced]... 26040.85 ms

Pi better before: True (21267.1 vs 26040.9)

Pj better after: True (19558.8 vs 24367.7)

Valid optimal switch: YES

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Query 3 — Switch S1 (EARLY)

Optimizer's switch: 1992-01-01 → 1992-01-02

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Searching for optimal switch...

Boundary [1998-08-02]: RT(Pi)=14093.6ms RT(Pj)=13814.6ms

[1992-01-02 ... 1995-04-18 ... 1998-08-02] RT(Pi)=16032.2 RT(Pj)=16550.7 → Pi better,
search RIGHT

[1995-04-18 ... 1996-12-09 ... 1998-08-02] RT(Pi)=15378.0 RT(Pj)=15233.8 → Pj better,
search LEFT

[1995-04-18 ... 1996-02-12 ... 1996-12-09] RT(Pi)=16559.4 RT(Pj)=17404.9 → Pi better,
search RIGHT

[1996-02-12 ... 1996-07-11 ... 1996-12-09] RT(Pi)=15734.6 RT(Pj)=16188.5 → Pi better,
search RIGHT

[1996-07-11 ... 1996-09-24 ... 1996-12-09] RT(Pi)=15967.0 RT(Pj)=16133.3 → Pi better,
search RIGHT

[1996-09-24 ... 1996-11-01 ... 1996-12-09] RT(Pi)=15008.2 RT(Pj)=15958.6 → Pi better,
search RIGHT

[1996-11-01 ... 1996-11-20 ... 1996-12-09] RT(Pi)=15356.6 RT(Pj)=15679.3 → Pi better,
search RIGHT

[1996-11-20 ... 1996-11-29 ... 1996-12-09] RT(Pi)=15329.0 RT(Pj)=18211.2 → Pi better,
search RIGHT

[1996-11-29 ... 1996-12-04 ... 1996-12-09] RT(Pi)=15553.8 RT(Pj)=16324.6 → Pi better,
search RIGHT

[1996-12-04 ... 1996-12-06 ... 1996-12-09] RT(Pi)=15962.5 RT(Pj)=15647.7 → Pj better,
search LEFT

[1996-12-04 ... 1996-12-05 ... 1996-12-06] RT(Pi)=16223.4 RT(Pj)=15976.4 → Pj better,
search LEFT

Optimal switch found: 1996-12-04 → 1996-12-05

Validating...

Measuring RT(Pi, d*i=1996-12-04)... 16007.11 ms

Measuring RT(Pj, d*j=1996-12-05)... 15926.46 ms

Measuring RT(Pi, d*j=1996-12-05) [forced]... 16470.02 ms

Measuring RT(Pj, d*i=1996-12-04) [forced]... 19536.78 ms

Pi better before: True (16007.1 vs 19536.8)

Pj better after: True (15926.5 vs 16470.0)

Valid optimal switch: YES

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Query 3 — Switch S2 (DELAYED)

Optimizer's switch: 1992-01-03 → 1992-01-04

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Searching for optimal switch...

Boundary [1992-01-01]: RT(Pi)=0.0ms RT(Pj)=0.1ms

[1992-01-01 ... 1992-01-02 ... 1992-01-03] RT(Pi)=139.5 RT(Pj)=98.5 → Pj better, search
LEFT

Optimal switch found: 1992-01-01 → 1992-01-02

Validating...

Measuring RT(Pi, d*i=1992-01-01)... 0.02 ms

Measuring RT(Pj, d*j=1992-01-02)... 102.43 ms

Measuring RT(Pi, d*j=1992-01-02) [forced]... 96.76 ms

Measuring RT(Pj, d*i=1992-01-01) [forced]... 0.02 ms

Pi better before: True (0.0 vs 0.0)

Pj better after: False (102.4 vs 96.8)

Valid optimal switch: NO (boundary case)

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Query 3 — Switch S3 (EARLY)

Optimizer's switch: 1992-01-06 → 1992-01-07

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Searching for optimal switch...

Boundary [1998-08-02]: RT(Pi)=16005.3ms RT(Pj)=15404.3ms

[1992-01-07 ... 1995-04-20 ... 1998-08-02] RT(Pi)=17642.2 RT(Pj)=22143.3 → Pi better,
search RIGHT

[1995-04-20 ... 1996-12-10 ... 1998-08-02] RT(Pi)=16458.4 RT(Pj)=16382.7 → Pj better,
search LEFT

[1995-04-20 ... 1996-02-14 ... 1996-12-10] RT(Pi)=16714.4 RT(Pj)=16526.2 → Pj better,
search LEFT

[1995-04-20 ... 1995-09-17 ... 1996-02-14] RT(Pi)=17654.2 RT(Pj)=17518.7 → Pj better,
search LEFT

[1995-04-20 ... 1995-07-04 ... 1995-09-17] RT(Pi)=17428.5 RT(Pj)=16607.9 → Pj better,
search LEFT

[1995-04-20 ... 1995-05-27 ... 1995-07-04] RT(Pi)=16120.1 RT(Pj)=20509.1 → Pi better,
search RIGHT

[1995-05-27 ... 1995-06-15 ... 1995-07-04] RT(Pi)=16852.5 RT(Pj)=16774.3 → Pj better,
search LEFT

[1995-05-27 ... 1995-06-05 ... 1995-06-15] RT(Pi)=16381.8 RT(Pj)=20729.5 → Pi better,
search RIGHT

[1995-06-05 ... 1995-06-10 ... 1995-06-15] RT(Pi)=16082.2 RT(Pj)=16556.0 → Pi better,
search RIGHT

[1995-06-10 ... 1995-06-12 ... 1995-06-15] RT(Pi)=16555.3 RT(Pj)=16623.5 → Pi better,
search RIGHT

[1995-06-12 ... 1995-06-13 ... 1995-06-15] RT(Pi)=16881.8 RT(Pj)=17706.7 → Pi better,
search RIGHT

[1995-06-13 ... 1995-06-14 ... 1995-06-15] RT(Pi)=17248.2 RT(Pj)=16845.1 → Pj better,
search LEFT

Optimal switch found: 1995-06-13 → 1995-06-14

Validating...

Measuring RT(Pi, d*i=1995-06-13)... 17492.56 ms

Measuring RT(Pj, d*j=1995-06-14)... 16864.17 ms

Measuring RT(Pi, d*j=1995-06-14) [forced]... 17275.39 ms

Measuring RT(Pj, d*i=1995-06-13) [forced]... 22370.94 ms

Pi better before: True (17492.6 vs 22370.9)

Pj better after: True (16864.2 vs 17275.4)

Valid optimal switch: YES

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Query 3 — Switch S4 (EARLY)

Optimizer's switch: 1992-01-18 → 1992-01-19

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Searching for optimal switch...

Boundary [1998-08-02]: RT(Pi)=13894.4ms RT(Pj)=13897.5ms

Pi always better in range — no crossover found

No crossover found — Pi remains better throughout range

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Query 3 — Switch S6 (EARLY)

Optimizer's switch: 1992-03-08 → 1992-03-09

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Searching for optimal switch...

Boundary [1998-08-02]: RT(Pi)=13953.9ms RT(Pj)=14016.1ms

Pi always better in range — no crossover found

No crossover found — Pi remains better throughout range

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Query 3 — Switch S7 (DELAYED)

Optimizer's switch: 1992-03-13 → 1992-03-14

=====

Searching for optimal switch...

Boundary [1992-01-01]: RT(Pi)=0.0ms RT(Pj)=0.0ms

Pj better even at data_start — optimal switch is at very beginning

Optimal switch found: 1992-01-01 → 1992-01-02

Validating...

Measuring RT(Pi, d*i=1992-01-01)... 0.01 ms

Measuring RT(Pj, d*j=1992-01-02)... 152.59 ms

Measuring RT(Pi, d*j=1992-01-02) [forced]... 100.60 ms

Measuring RT(Pj, d*i=1992-01-01) [forced]... 0.02 ms

Pi better before: True (0.0 vs 0.0)

Pj better after: False (152.6 vs 100.6)

Valid optimal switch: NO (boundary case)